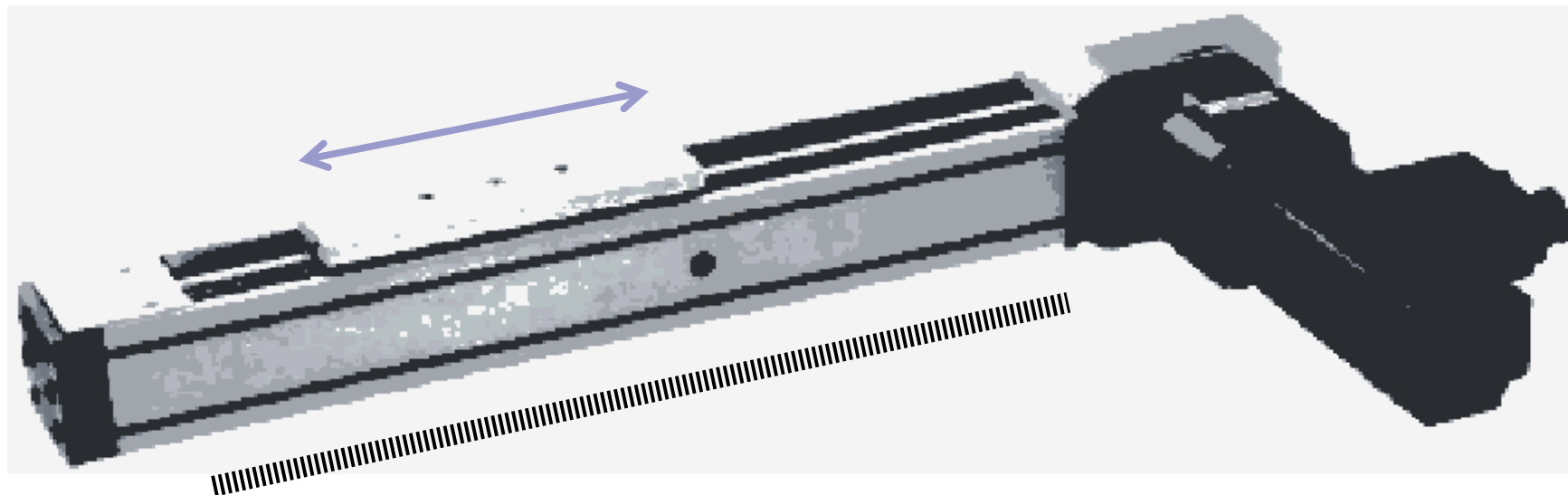
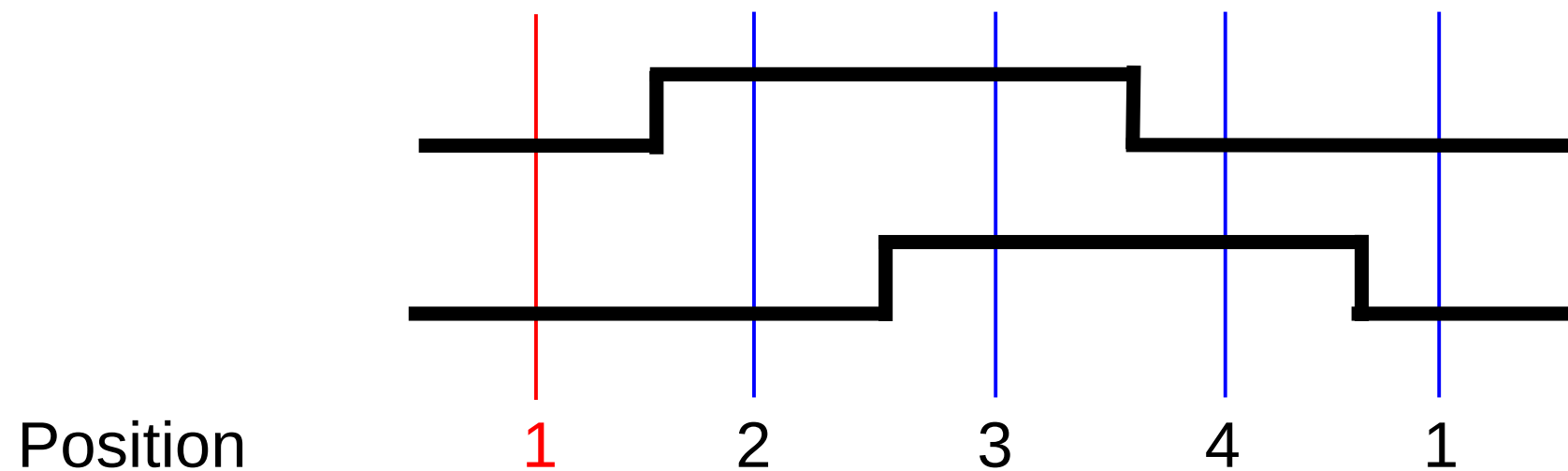


Zustandsgraph am Beispiel Linearbewegung



Zustandsgraph am Beispiel Linearbewegung

Doppelpuls erkennen



Zustand 1 : Position 1 , Richtung ->

Zustand 2 : Position 2 , Richtung ->

Zustand 3 : Position 3 , Richtung ->

Zustand 4 : Position 4 , Richtung ->

Zustand 5 : Position 1 , Richtung <-

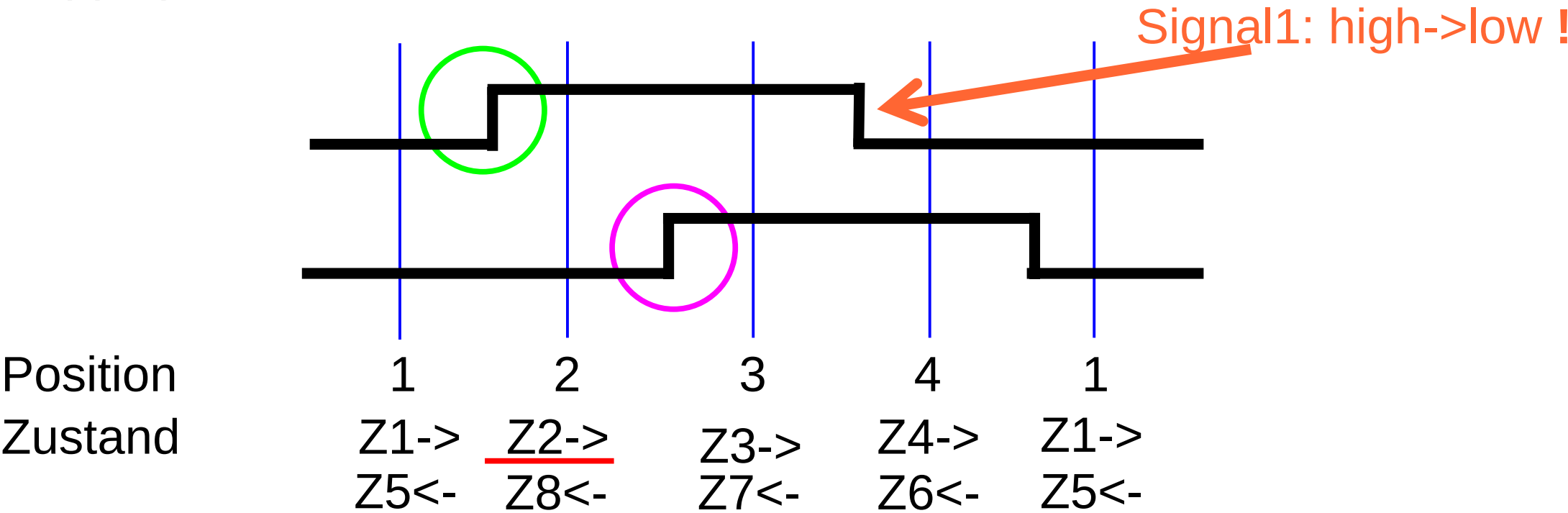
Zustand 6 : Position 4 , Richtung <-

Zustand 7 : Position 3 , Richtung <-

Zustand 8 : Position 2 , Richtung <-

Zustandsgraph am Beispiel Linearbewegung

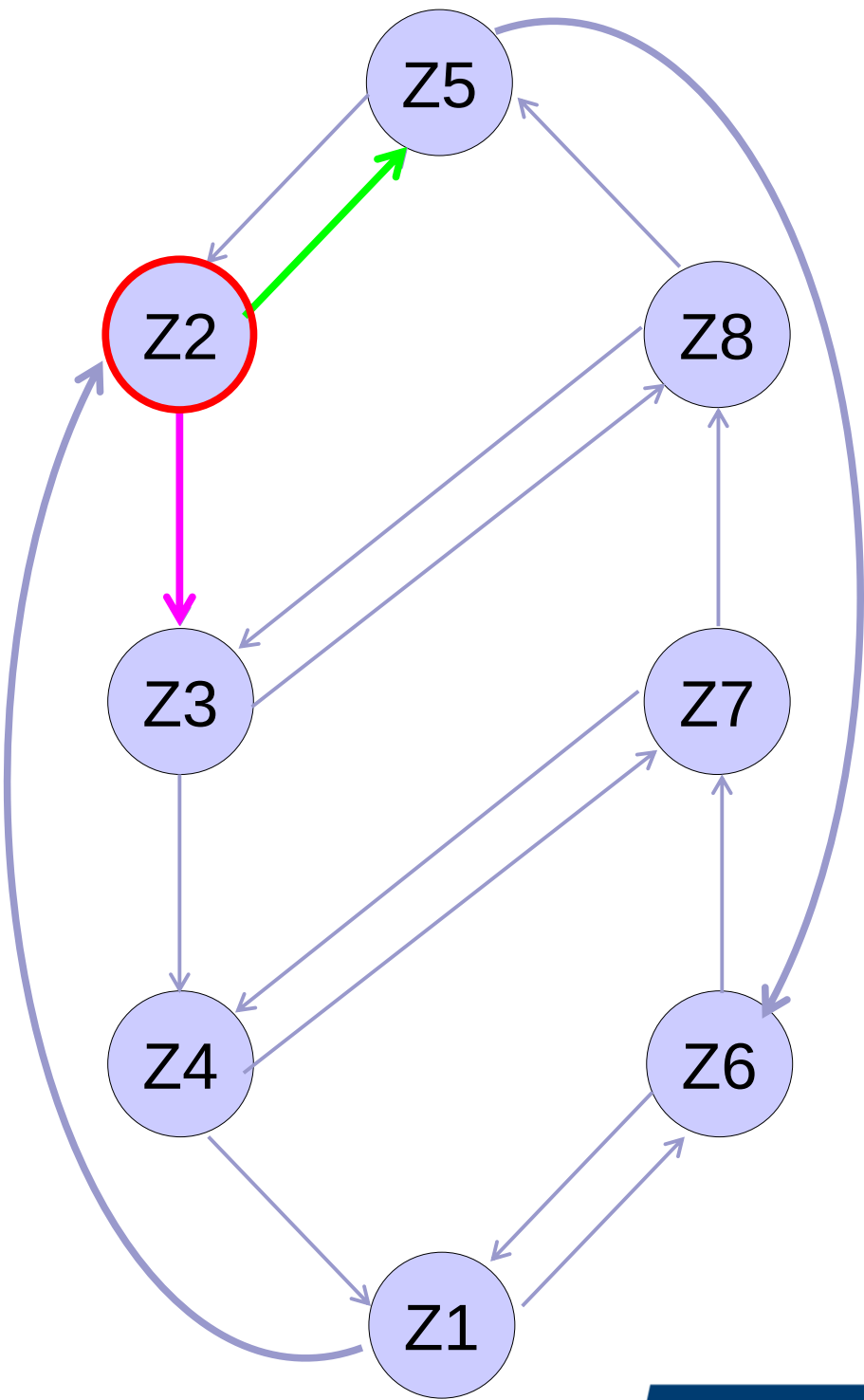
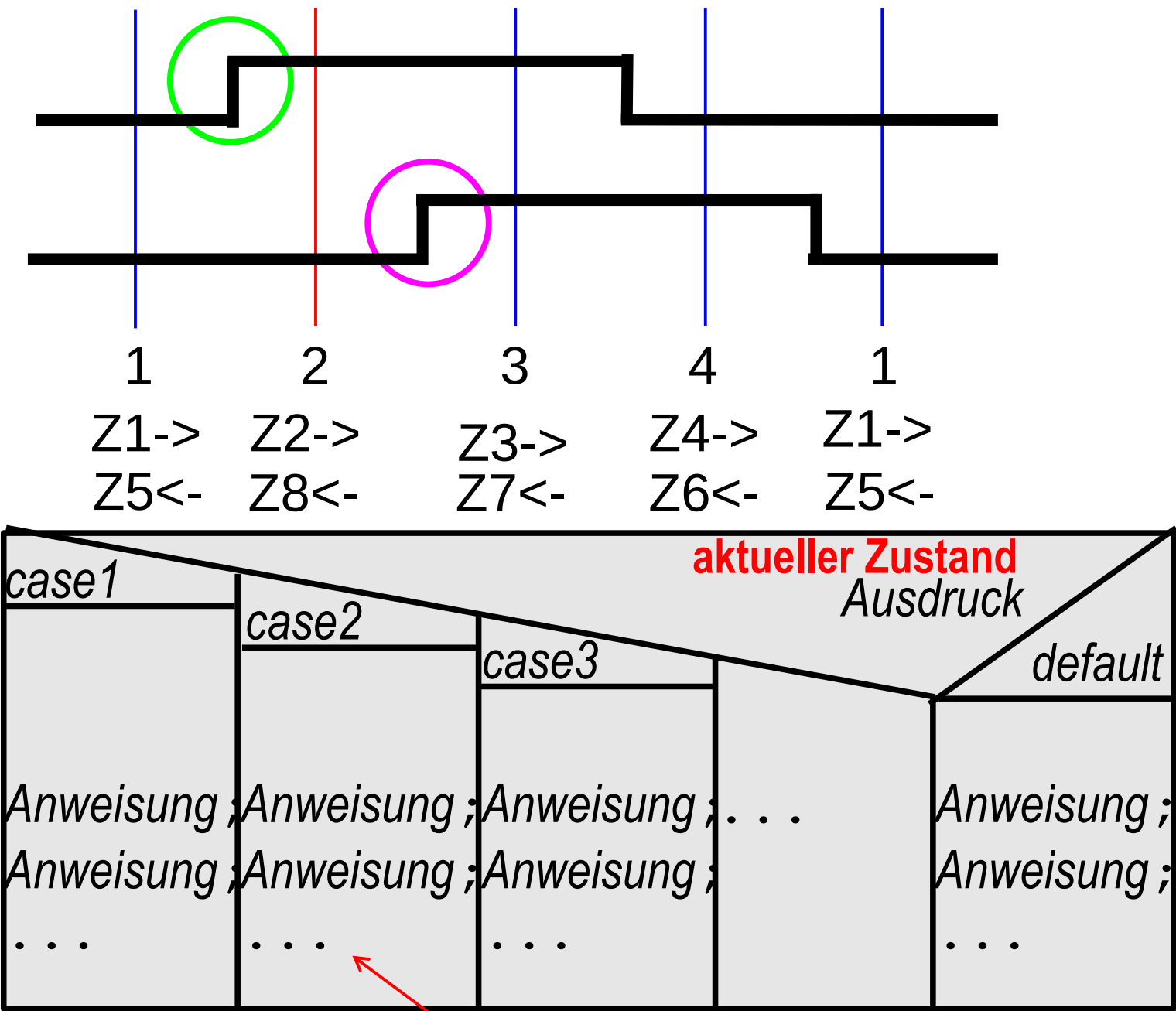
Doppelpuls erkennen



Aktueller Zustand	Ereignis	Folgezustand
...	...	...
<u>Zustand 2</u>	<u>Signal2: low->high</u>	<u>Zustand 3</u>
	<u>Signal1: high->low</u>	<u>Zustand 5</u>
...	...	...

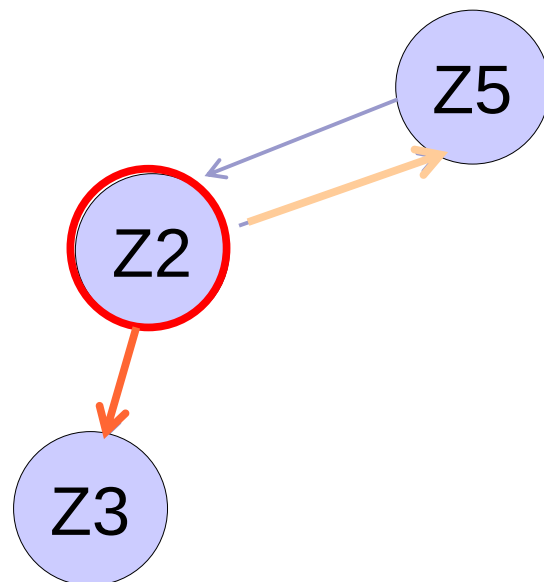
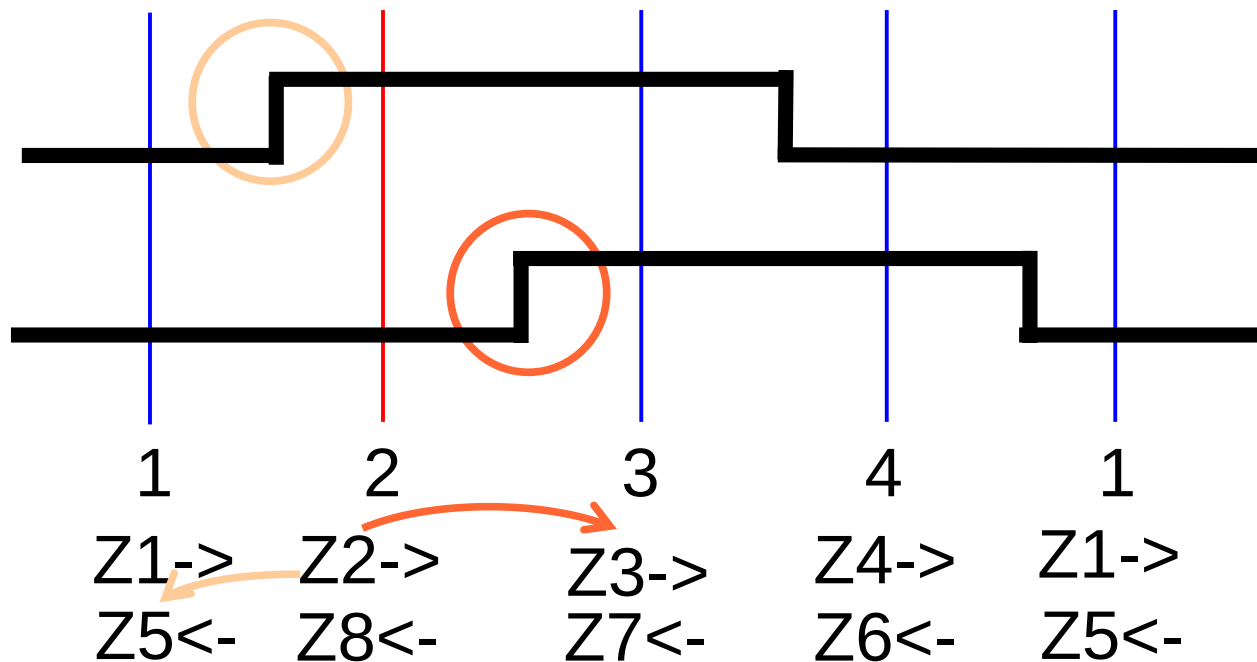
Zustandsgraph am Beispiel Linearbewegung

Doppelpuls erkennen



Zustandsgraph am Beispiel Linearbewegung

Doppelpuls erkennen

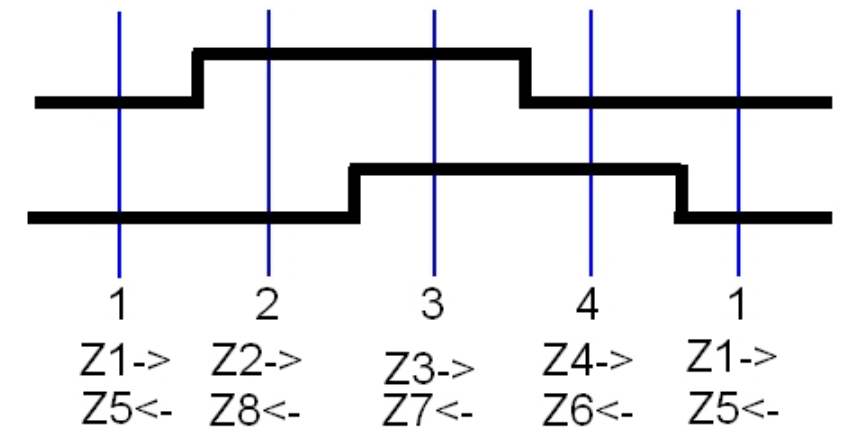


```
...  
#define Z1 1  
#define Z2 2  
#define Z3 3  
...  
switch (state) ← aktueller Zustand  
{  
    ...  
    //channel A: high, channel B: low  
    case Z2:  
        ... Ereignis führt zu Folgezustand  
        if (channel_b == HIGH)  
        {  
            state = Z3;  
        }  
        if (channel_a == LOW)  
        {  
            state = Z5;  
        }  
        break; Ereignis führt zu Folgezustand  
    //channel A: high, channel B: high  
    case Z3:  
        ...  
        break;  
    ...  
}
```

```

...
for (;;)
{
    ... // sensoren einlesen
    switch (state)
    {
        ...
        //channel A: high, channel B: low
        case Z2:
            if (channel_b == HIGH)
            {
                state = Z3; position++; richtung = 1;
            }
            if (channel_a == LOW)
            {
                state = Z5; position--; richtung = 0;
            }
            break;
        //channel A: high, channel B: high
        case Z3:
            if (channel_b == LOW)
            {
                state = Z8; position--; richtung = 0;
            }
            if (channel_a == LOW)
            {
                state = Z4; position++; richtung = 1;
            }
            break;
        ...
    }
}

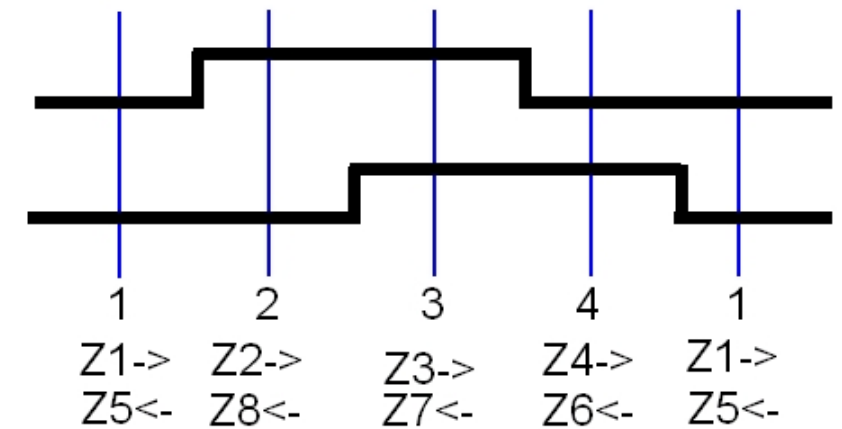
```



```

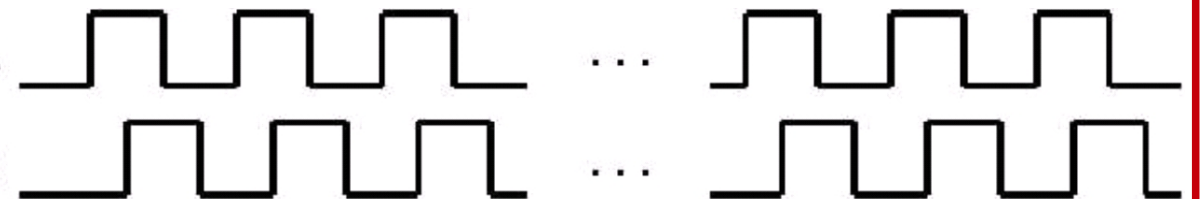
...
for (;;)
{
    ... // sensoren einlesen
    switch (state)
    {
        ...
        //channel A: high, channel B: low
        case Z2:
            if (channel_b == HIGH)
            {
                state = Z3; position++; richtung = 1;
            }
            if (channel_a == LOW)
            {
                state = Z5; position--; richtung = 0;
            }
            break;
        //channel A: high, channel B: high
        case Z3:
            if (channel_b == LOW)
            {
                state = Z8; position--; richtung = 0;
            }
            if (channel_a == LOW)
            {
                state = Z4; position++; richtung = 1;
            }
            break;
        ...
    }
}

```



Kanal A

Kanal B




```

...
for (;;)
{
    ... // sensoren einlesen
    switch (state)
    {

```

```

        ...
        //channel A: high, channel B: low
        case Z2:
            if (channel_b == HIGH)
            {
                state = Z3; position++; richtung = 1;
            }
            if (channel_a == LOW)
            {
                state = Z5; position--; richtung = 0;
            }
            break;

```

```

        //channel A: high, channel B: high
        case Z3:
            if (channel_b == LOW)
            {
                state = Z8; position--; richtung = 0;
            }
            if (channel_a == LOW)
            {
                state = Z4; position++; richtung = 1;
            }
            break;
        ...
    }
}

```

